Speaker–microphone bench verification

SONIC Auditory Benchmarking 23 October 2026

Established. Simultaneous speaker playback and microphone recording work. Two timing runs established an approximately **38.3 ms application roundtrip** in the tested program and wired configuration.

Setup. Computer/Core Audio at 96 kHz → Audient iD14 → Edifier speaker → Earthworks M50 → Audient input 1 → program. The operator reported 10 cm speaker–microphone separation; it was not remeasured.

SONIC sounds. We played and recorded one reconstructed Sheep E block using the paper's exact 14-frequency alphabet (kHz):

$$\{2,\ 4,\ 6,\ 8,\ 9,\ 10.5,\ 17,\ 18.2,\ 19.5,\ 20.9,\ 22.4,\ 24,\ 25.8,\ 27\}.$$

The block contains 8,400 consecutive 10 ms zero-phase sine pips, with 2 ms ramps at each end. The original tone order and ramp shape were unavailable: our seeded order and raised-cosine ramps are declared reconstruction choices. Output was 96 kHz; the paper used 192 kHz.

Timing result. Forty nonperiodic 100 Hz–20 kHz sweeps were matched from the exact output array to microphone input 1, across two separately started full-duplex streams.

Run	Accepted	Median (ms)	Observed range (ms)
1	20/20	38.250	38.240–38.250
2	20/20	38.375	38.365–38.375

All probes passed signal and transport checks; there was no clipping or callback warning. Run medians differed by 0.125 ms. Direct delay estimates from the SONIC sequence remained cycle-ambiguous; **38.3 ms is the separate sweep-based result**.

Scope. This establishes short-run repeatability, including playback/recording buffers, conversion and the acoustic chain. It does not establish absolute accuracy, isolated speaker delay, calibrated SPL, ultrasonic coverage, or another program's latency.

Records: timing measurements, SONIC playback. Source: SONIC v1, Methods 6.1–6.2.